# NEURAL NETWORKS · AUTOENCODER TUTORIAL
# What Does an Autoencoder Remember?

*Investigating Latent Dimension and Quality of Reconstruction*

| **Technique:** Fully Connected Autoencoder | **Dataset**: Fashion-MNIST | **Focus:** Latent Dimension | **Dims tested:** 2, 8, 16, 32, 64 |
|---|---|---|---|
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## OBJECTIVE

This tutorial explores one of the most impactful design choices in the construction of an autoencoder how many dimensions is the bottleneck supposed to be? Training five otherwise identical models on Fashion-MNIST with latent dimensions of 2, 8, 16, 32 and 64, and holding all other hyperparameters constant, we are isolating precisely the effect the bottleneck size has on reconstruction quality.

*KEY QUESTION*
*How little information can a model compress into before it forgets too much to reconstruct anything useful?*

## INTRODUCTION

An autoencoder is a neural network that is trained to reduce its input to a small representation and then it is reconstructed based on the small representation. It has two parts, the encoder, which encodes input information into a low-dimensional latent code, and the decoder, which recodes the original input based on the latent code. The latent space or bottleneck is known as the compressed representation. Because the network must reproduce its input through this narrow passage, it is forced to learn the most important features of the data and discard noise and redundancy (Hinton, G.E. et al 2006, Goodfellow, et al 2016)

Autoencoders are not supervised: they do not require any labels in training. They find application in dimensionality reduction, anomaly detection, denoising, feature learning, and as components of generative models. (Hinton, G.E.et al. R. (2006) Vincent, P. et al. 2010)

*KEY INSIGHT*
*The bottleneck forces the model to be selective. It cannot afford to memorise everything; it must learn what matters most about the data.*

## BUILDING AN INTUITION

Think of the autoencoder as a postal system where you must describe a photograph using only a fixed number of words. With 2 words, you might manage “grey shoe”, broad shape, no detail The laces, the sole pattern, the stitching, can all be described with 32 words. The recipient is able to build something that is much nearer to the original.

The latent dimension is that word limit. Changing it changes what the model can say about the image and therefore what can be recovered. This trade off is rooted in information theory: the bottleneck acts as a lossy channel. The encoder has to decide on what details to save and anything that it

drops cannot be reclaimed by the decoder (Cover, T.M. and Thomas, J.A. 2006). The higher the bottleneck, the higher the channel capacity; the smaller, the more difficult decisions must be made .

**ARCHITECTURE**

The autoencoder used in this tutorial is fully connected (dense layers only). The architecture is symmetric:

```
Input (784)  →  256  →  128  →  [latent_dim]  →  128  →  256  →  Output (784)

Activations: ReLU (hidden layers)  |  Sigmoid (output layer)
Loss: Mean Squared Error (MSE)  |  Optimiser: Adam (lr=1e-3)
```

ReLU activations are present in the hidden layers allowing the model to learn non-linear transformations. The sigmoid of the final output layer rescales pixel values to [0, 1], which is similar to the normalised input. The latent dimension d changes in experiments and the other components remain constant making the comparison controlled. PyTorch implementation is concise:

```
class Autoencoder(nn.Module):
    def __init__(self, latent_dim):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Linear(784, 256), nn.ReLU(),
            nn.Linear(256, 128), nn.ReLU(),
            nn.Linear(128, latent_dim)
        )
        self.decoder = nn.Sequential(
            nn.Linear(latent_dim, 128), nn.ReLU(),
            nn.Linear(128, 256),        nn.ReLU(),
            nn.Linear(256, 784),        nn.Sigmoid()
        )
    def forward(self, x):
        return self.decoder(self.encoder(x.view(-1, 784)))
```

**DATASET: FASHION-MNIST**

Fashion-MNIST (Xiao, H., Rasul, K. and Vollgraf, R. 2017) is a collection of 10 categories of clothing, each containing 70,000 greyscale 28x28 images: T-shirts, trousers, pullovers, dresses, coats, sandals, shirts, sneakers, bags and ankle boots.

It was selected in comparison to regular MNIST due to the existence of a wider range of structure, edges, textures, and silhouette in clothing items.

Reconstruction quality differences between latent dimensions more visually apparent.

Most importantly, autoencoders are not trained using class labels. The task is just to rebuild the input as closely as possible using the bottleneck. Class labels are only used afterwards to colour the latent space visualisation.

*CONTROLLED EXPERIMENT*

*Only the latent dimension changes. Architecture depth, hidden layer sizes, optimiser, learning rate, epochs, and batch size are identical across all five models. Any difference in performance is therefore attributable solely to the bottleneck size.*

## RESULTS

### Reconstruction Loss

The table below shows the final test reconstruction loss (MSE) for each model. Lower values indicate better reconstruction, and for SSIM higher values indicate better reconstruction.

| Latent Dim | Final Test Lose(MSE) | Compression | SSIM | Assesment |
|---|---|---|---|---|
| 2 | 0.029981 | 392 :1 | 0.5351 | Poor (Severe over-compression) |
| 8 | 0.017517 | 98 : 1 | 0.6570 | Good (Large gain over dim=2) |
| 16 | 0.016119 | 49 : 1 | 0.6726 | Best in this experiment |
| 32 | 0.016385 | 24 : 1 | 0.6666 | Good reconstruction |
| 64 | 0.016459 | 12 : 1 | 0.6642 | Possible mild overfit |

The most striking finding is the cliff between dim=2 and dim=8. Loss nearly halves with just six extra dimensions, confirming that 2 dimensions is genuinely insufficient to encode even the coarsest structure of Fashion-MNIST images.

Absolute values differ slightly between 8, 16, 32 and 64. The fact that it was slightly higher at dim=64 is perhaps due to a small overfitting effect, but this is very slight. This trend is the phenomenon of diminishing returns: as soon as the latent space has grown large enough to represent primary visual structure, the extra dimensions obtain smaller payoffs.

## VISUAL RECONSTRUCTIONS

Quantitative loss is one lens; visual inspection is another. At dim=2, the clothing types are hardly differentiated, a coat and a t-shirt may be mixed up easily. The pictures are dark and grey. Silhouettes, relative brightness, and large, major structural features are retained at the dim=32. Even the finest reconstructions are not completely sharp due to the inability to utilize spatial locality as a convolutional autoencoder would do (Goodfellow, I., Bengio, Y. and Courville, A. 2016).

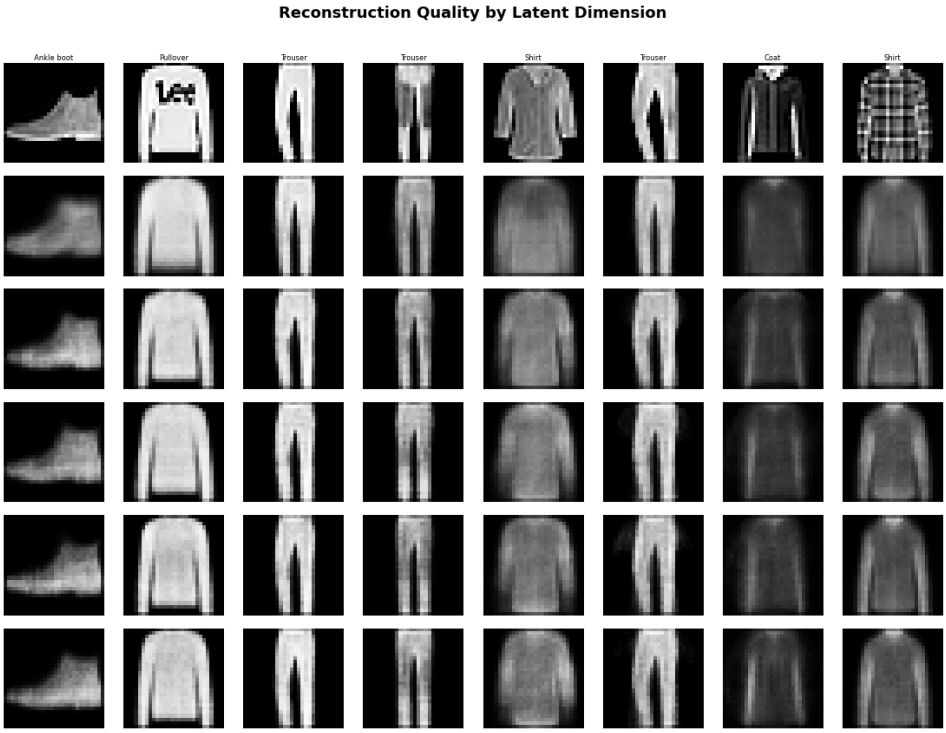


*Figure 1. Originals and reconstructions of Fashion-MNIST images. Latent-2 model reconstructions are less detailed and blurry whereas bigger latent dimensions retain more visual structure.*

## SSIM: PERCEPTUAL QUALITY

Besides MSE, we also calculate Structural Similarity Index (SSIM) (Wang, Z. et al. 2004) which is a perceptual quality measure. SSIM takes into consideration luminance, contrast, and structure, which is more a reflection of what is perceived by humans as to the quality of visual perception. SSIM is between 0 and 1 (the higher the better). Findings are similar to MSE, with dim=2 lowest, followed by plateau at dim=8-16.

## THE 2D LATENT SPACE

When the latent dimension goes to be precisely equal to 2, all test images can be represented as a point in 2D space, providing a direct view into what the model has trained. Class labels were never presented to the autoencoder, but the latent space is loosely clustered by category of clothing. The same objects (bags, sneakers, ankle boots) are forced to the same parts of the space.

This emergent structure arises because items that look alike are compressed similarly. The autoencoder has learned an internal geometry of the data that reflects genuine visual similarity Hinton, G.E. and Salakhutdinov, R.R. 2006, Vincent, P. et al. (2010). This latent structure may be re-purposed to downstream consists of nearest-neighbour retrieval or anomaly detection, never having labelled a single training example.

*ACCESSIBILITY NOTE*

*The latent space scatter plot uses both colour and distinct marker shapes (circles, squares, triangles, etc.) for each clothing category, ensuring the figure is interpretable by colour-blind readers.*

## LATENT SPACE INTERPOLATION

Linear interpolation is a potent method of exploration in latent space: pick two images, encode them, linearly interpolate the latent vectors of the two images, and decode every interpolation step. In case the latent space is smooth and well structured the decoded images must move gradually between the two inputs. There are discontinuities or gaps in the latent representation that are uncovered by abrupt jumps.The notebook (Section 9) demonstrates this for the dim=32 model, interpolating between an ankle boot and a coat. The smooth transition confirms that the model has learned a meaningful, continuous representation of the input space, not merely a lookup table.

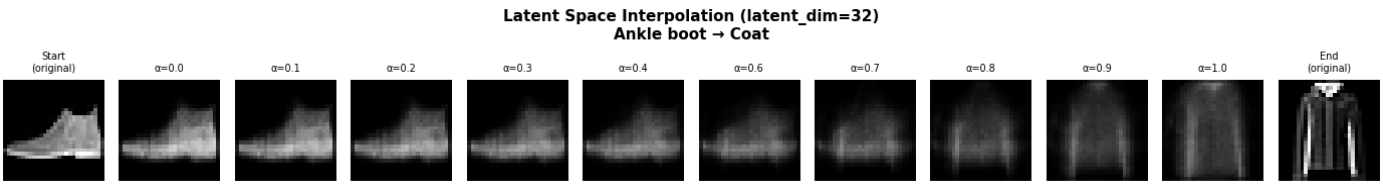


*Figure 2. dim=32 model, interpolating between an ankle boot and a coat.*

## SELECTING A LATENT DIMENSION IN PRACTICE

The results above suggest practical guidance. The key question is: what is your bottleneck for?

- For direct 2D/3D visualisation only. Reconstruction will suffer. Best for exploratory analysis of data structure.2–3 dimensions:
- Good balance of compression and quality for simple image datasets. The ‘elbow’ in this experiment lies in this range.8–16 dimensions:

- Better reconstruction at the cost of weaker compression. Suitable when downstream tasks require rich feature representations.32–64 dimensions:
- Diminishing returns. Useful for complex or high-resolution inputs. Consider a Variational Autoencoder (VAE) (Kingma, D.P. and Welling, M. 2014) at this scale.128+:

A practical strategy is to train a sweep (as done here), plot loss as a function of dimension, and choose the ‘elbow’, the point where additional dimensions yield only marginal improvement.

*BEYOND THIS TUTORIAL*
*A Variational Autoencoder (VAE) (Kingma, D.P. and Welling, M. 2014) adds a regularisation term encouraging the latent space to follow a smooth, continuous distribution. This makes interpolation and generation meaningful; you can sample new points and produce plausible new images. VAEs typically use their bottleneck more efficiently than standard autoencoders.*

**ETHICAL CONSIDERATIONS**

Autoencoders are effective exactly due to the fact that they are unsupervised learners. This, however, implies that the learned latent space will capture all the biases in the training data in a manner that there is no explicit mechanism to uncover or rectify them (Goodfellow, I., Bengio, Y. and Courville, A. 2016).

When trained on biased data (e.g. medical images which do not represent all groups), an autoencoder will recreate its images and latent representations in a biased way. Anomaly detection algorithms constructed based on such autoencoders can declare underrepresented groups of people anomalies not because they are actually anomalous, but due to the lack of examples of them in the model.

Additionally, reconstruction loss alone is not a complete measure of representation quality A model can also attain low MSE by maintaining coarse structure and removing fine grained detail which could be important to downstream applications, like subtle lesion detail in medical imaging.
Practitioners ought to: audit training data to detect representations gaps; validate latent representations on out-of-reconstruction tasks; and that they should be especially careful when making life-or-death choices with high stakes, like in the medical sector, security, or financial services.

**CONCLUSION**

In this tutorial, the latent dimension has been demonstrated as a fundamental architectural choice that shapes what was kept in memory by an autoencoder, what was forgotten, and the utility of the representations.
Key findings from training five models on Fashion-MNIST:

- Latent dim=2 causes severe information loss; reconstruction nearly doubles in MSE vs dim=8
- The largest single gain occurs between dim=2 and dim=8
- Beyond dim=8, gains diminish rapidly; the elbow lies between 8 and 16
- Dim=16 achieved the lowest test loss in this experiment
- Even without labels, the 2D latent space shows emergent clustering by clothing category

## TEACHING CHOICES AND ACCESSIBILITY

This tutorial is user friendly:

- All scatter plots use both colour and distinct marker shapes to support colour-blind readers
- Figure captions are descriptive enough to convey findings without requiring visual inspection
- Code blocks use syntax that is readable in screen-reader-compatible formats

## REFERENCES


Cover, T.M. and Thomas, J.A. (2006) Elements of Information Theory. 2nd edn. Wiley-Interscience.

Goodfellow, I., Bengio, Y. and Courville, A. (2016) Deep Learning. MIT Press. https://www.deeplearningbook.org

Hinton, G.E. and Salakhutdinov, R.R. (2006) 'Reducing the dimensionality of data with neural networks', Science, 313(5786), pp. 504–507. https://doi.org/10.1126/science.1127647

Kingma, D.P. and Welling, M. (2014) Auto-encoding variational bayes. arXiv:1312.6114. https://arxiv.org/abs/1312.6114

PyTorch (2024) Documentation. https://pytorch.org/docs (Accessed: March 2025).

Vincent, P. et al. (2010) 'Stacked denoising autoencoders', JMLR, 11, pp. 3371–3408.

Wang, Z. et al. (2004) 'Image quality assessment: from error visibility to structural similarity', IEEE TIP, 13(4), pp. 600–612.

Xiao, H., Rasul, K. and Vollgraf, R. (2017) Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms. arXiv:1708.07747. https://arxiv.org/abs/1708.07747